

Shubham Kumar

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Backend Engineer with 5 years of experience architecting **distributed systems** and **event-driven architectures**. Expert in designing high-throughput production systems serving **2M+ users** and processing **25K+ events/sec** using **Python, Node.js, and PostgreSQL**. Proven track record in optimizing **microservices performance**, implementing **vector-based ranking**, and mentoring engineering teams to deliver scalable, low-latency solutions.

TECHNICAL SKILLS

- **Languages:** Python, C++, JavaScript, Go (learning)
- **Backend:** Django, Node.js
- **Databases:** PostgreSQL, Redis, Elasticsearch
- **Architecture:** Microservices, Event-Driven Systems, Distributed Systems
- **Messaging/Async:** Celery, Redis Pub/Sub, WebSockets, Background Workers
- **Cloud/Infra:** AWS EC2, Docker, Nginx, Traefik, S3, SQS, Lambda
- **System Design:** HLD, LLD, API Design
- **Engineering:** SOLID Principles, Design Patterns, Performance Optimization

WORK EXPERIENCE

Under25 - Collective Artists Network | Bangalore, Karnataka

Senior Backend Engineer | September 2024 - Present

- Architected a personalized feed ranking system using vector embeddings and community detection to surface relevant content based on user interactions and profile signals, increasing platform engagement by **30%**.
- Built a high-throughput leaderboard system for **2M+ users**, processing burst activity streams of **~25K events/sec** through an event-driven pipeline to ensure accurate and consistent rankings at scale.
- Designed a scalable RBAC-based authorization and moderation framework enabling hierarchical permissions for internal teams and community moderators to securely govern content and user activities across platform spaces.
- Improved database performance by analyzing execution plans and optimizing slow queries with indexing strategies and query refactoring, reducing database load and lowering latency for high-traffic APIs.
- Mentored 2 junior engineers through backend architecture guidance, conducted backend PR reviews, and enforced performance and code quality standards across services.
- Tech Stack: Python, Django, PostgreSQL, AWS EC2, Nginx, Elasticsearch, Redis (ElastiCache), Celery

Independent Backend Engineer (Freelance)

November 2023 - August 2024

- Architected a microservices-based system for an on-demand 3D printing platform supporting CAD/STL uploads, asynchronous processing, and malware scanning, with services orchestrated behind Traefik for routing and zero-downtime deployments. | shap3d.in
- Engineered a real-time trading data pipeline ingesting market streams for 500 stocks from Zerodha Kite WebSocket APIs and distributing events via Redis Pub/Sub to storage services and real-time trading clients.
- Developed backend infrastructure for a yoga wellness platform, implementing content publishing, workshop registration, and scheduled workflows using Celery with Redis caching and ETA-based background task scheduling. | theyogicalgorithm.in
- Built service for a healthcare platform implementing transactional appointment scheduling with calendar slot locking, along with automated email and SMS notifications via MSG91. | tinystepspediatrics.com
- Tech Stack: Python, Django, PostgreSQL, Redis, Celery, WebSockets, Docker, Prometheus, Traefik

Radisys Corporation | Bangalore, Karnataka

Senior Software Engineer, 5G Platform Team | December 2022 - November 2023

- Optimized server boot sequence, reducing startup time by 50+ seconds and decreasing binary size by 72%.
- Enhanced reliability of system monitoring tools by investigating crash dump files, resolving memory leaks, and implementing compression strategies to reduce dump size and improve production fault diagnostics.
- Increased test coverage by **35%** by expanding automated test suites and integrating build pipeline validation with static analysis tools.
- Led internal initiatives across a 5-member team, reviewing changes for core platform repositories.
- Tech Stack: C/C++, RedHat OS, Boost, DPDK, GCC, Shell Scripting, Python

Software Engineer, 5G Platform Team | July 2021 - December 2022

- Resolved 15+ critical bugs in System Monitor, eliminating false positives affecting 3 production deployments.
- Added 40+ unit and integration test cases for core monitoring tools.
- Coordinated with the OAM team to validate rollback scenarios across 3 major release cycles.
- Tech Stack: C/C++, Bash, Object-Oriented Programming, Python

Think201 | Bangalore, Karnataka (Remote)

Software Engineer Intern, Web Application Backend Team | September 2020 - November 2020

- Designed database schemas and implemented RESTful API endpoints for multiple web applications using Node.js and PostgreSQL under mentor-guided code reviews.
- Collaborated with frontend engineers across sprint cycles to deliver backend features, validating APIs through Postman-based testing.
- Tech Stack: Typescript, Node.js, PostgreSQL, Swagger API, Postman, Sequelize

EDUCATION

National Institute of Technology Karnataka | Surathkal, Karnataka

M.Tech(Research) in Computer Science & Engineering - Information Security | August 2019 - December 2022

Lovely Professional University | Phagwara, Punjab

B.Tech in Computer Science & Engineering | September 2015 - June 2019

ACHIEVEMENTS

- Published research papers on **Mirai malware analysis, and DDoS prevention in IoT.**